PHP Lecture 2
Lab Exercises - Setup and Instructions

Prerequisite: Ensure XAMPP (or MAMP) is installed and running (Apache and MySQL services should be started).

Setup Instructions:

1. **Download** the lecture_2.zip file from the LMS site.
1. **Extract the ZIP file.** This will create a folder named lecture_2.
2. **Move the folder to your server's root directory:**
    - **For XAMPP (Windows):** Move the lecture_2 folder to C:\xampp\htdocs\
    - **For MAMP (Mac):** Move the lecture_2 folder to /Applications/MAMP/htdocs/
3. **Access the Labs:** Open your web browser and navigate to the following URL to see the list of files:
   http://localhost/lecture_2/

You are now ready to begin. For each lab below, you will open the corresponding file, edit it with a code editor (like VS Code), save your changes, and then refresh its page in the browser to see the result.

Lab 1: Variables (lab1_variables.php**)**

Objective: Learn to declare variables and output their values.

Steps:

1. Open lab1_variables.php in your code editor
2. Locate the line: $name = 'Ivy';
3. Change 'Ivy' to your first name
4. Find the line: $price = 5;
5. Change 5 to 2
6. Save the file
7. View the result in your browser at: http://localhost/lecture_2/lab1_variables.php

Expected Change: The page should now show your name and the updated price.

Lab 2: Updating Variables (lab2_updating-variables.php**)**

Objective: Understand variable reassignment.

Steps:

1. Open lab2_updating-variables.php
2. Find the second assignment: $name = 'Ivy';
3. Change 'Ivy' to your name
4. Add a new line immediately after this assignment
5. On the new line, add: $name = 'Alex';
6. Save the file
7. Refresh the browser

Expected Change: The page will show "Alex" instead of your name, demonstrating the last assignment takes effect.

Lab 3: Associative Arrays (lab3_associative-arrays.php**)**

Objective: Work with associative arrays.

Steps:

1. Open lab3_associative-arrays.php
2. Find the $nutrition array
3. Change the values:
    - 'fat' => 42
    - 'sugar' => 60
    - 'salt' => 3.5
4. Add a new element after 'salt': 'protein' => 2.6,
5. In the HTML section, add this new line before </body>:

    Protein: <?php echo $nutrition['protein']; ?>%
6. Save and refresh

Expected Change: Updated nutrition values with a new Protein line.

Lab 4: Indexed Arrays (lab4_indexed-arrays.php**)**

Objective: Work with indexed arrays.

Steps:

1. Open lab4_indexed-arrays.php
2. Find the $best_sellers array
3. Add 'Licorice' after 'Fudge' (remember the comma)
4. In the HTML <ul> section, add these two lines after the existing <li> items:

   <?php echo $best_sellers[3]; ?>

   <?php echo $best_sellers[4]; ?>
5. Save and refresh

Expected Change: Five items in the best sellers list, including Licorice.

Lab 5: Updating Arrays (lab5_updating-arrays.php**)**

Objective: Modify existing arrays.

Steps:

1. Open lab5_updating-arrays.php
2. After the line: $nutrition['fiber'] = 2.1;
3. Add: $nutrition['protein'] = 7.3;
4. In the HTML section, add this line before </body>:

   Protein: <?php echo $nutrition['protein']; ?>%
5. Save and refresh

Expected Change: The nutrition display now includes Protein at 7.3%.

Lab 6: Multidimensional Arrays (lab6_multidimensional-arrays.php**)**

Objective: Work with nested arrays.

Steps:

1. Open lab6_multidimensional-arrays.php
2. Find the $offers array
3. Add this new array as the last element:

   ['name' => 'Chocolate', 'price' => 2, 'stock' => 83],
4. In the HTML section, add this line after the existing tags:

<?php echo $offers[3]['name']; ?> - $<?php echo $offers[3]['price']; ?>

5. Save and refresh

Expected Change: Four product offers displayed, including Chocolate.

Lab 7: Echo Shorthand (lab7_echo-shorthand.php**)**

Objective: Practice shorthand echo syntax.

Steps:

1. Open lab7_echo-shorthand.php
2. Change $name = 'Ivy'; to your name
3. In the $favorites array, add your favorite candy at the beginning:

'Your Favorite Candy',

4. Save and refresh

Expected Change: Page shows your name and your favorite candy.

Lab 8: Arithmetic Operators (lab8_arithmetic-operators.php**)**

Objective: Practice arithmetic operations.

Steps:

1. Open lab8_arithmetic-operators.php
2. Change $items = 3; to $items = 5;
3. Change $cost = 5; to $cost = 4;
4. Save and refresh

Expected Change: All calculated values (subtotal, tax, total) update automatically.

Lab 9: String Operator (lab9_string-operator.php**)**

Objective: Practice string concatenation.

Steps:

1. Open lab9_string-operator.php
2. Change $name = 'Ivy'; to your name
3. Comment out the current $message line by adding // at the beginning
4. Add this new line below it:

   $message = "$prefix, $name";

5. Save and refresh

Expected Change: The message appears identical, demonstrating two ways to combine strings.

Lab 10: Comparison Operators (lab10_comparison-operators.php**)**

Objective: Understand comparison operations.

Steps:

1. Open lab10_comparison-operators.php
2. Change $stock = 5; to $stock = 8;
3. Change $wanted = 8; to $wanted = 5;
4. Save and refresh

Expected Change: The "Can buy:" value changes from blank to "1" (true).

Lab 11: Logical Operators (lab11_logical-operators.php**)**

Objective: Understand logical operations.

Steps:

1. Open lab11_logical-operators.php
2. Change $stock = 5; to $stock = 3;
3. Change $wanted = 3; to $wanted = 5;
4. Save and refresh

Expected Change: The "Can buy:" value changes from "1" to blank (false).

Note: For all exercises, remember to save your changes in the code editor and refresh your browser to see the results. If you encounter errors, check your syntax carefully, especially commas in arrays and proper closing of PHP tags.